open data | data on request | closed data | preprint | reanalysis on open data | our reproduction

(a) Maximum vocabulary size used in the 32 language-BCI studies

vocabulary size

71,400 word forms (Brysbaert et al., 2016)
42,000 lemmas (Brysbaert et al., 2016)
11,100 word families (Brysbaert et al., 2016)

6 | 8 | 5–10 | 50 | 178 | 1,024 | 1,024 | 1,152 | 100k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 130k | 130k | open | open | open | open

Angrick et al., 2024 (CCD1) | Wandelt et al., 2024 (P1/P2) | Brosler et al., 2026 (B1r/B3/B6) | Moses et al., 2021 (B1) | Silva et al., 2024b (B1) | Metzger et al., 2023 (B3) | Littlejohn et al., 2025 (B3) | Metzger et al., 2022 (B1) | Feghhi et al., 2026 (T12/T15) | Willett et al., 2023 (T12) | Card et al., 2024 (T15) | Feng et al., 2024 (T12) | Kunz et al., 2025 (T12/T15/T16) | Li et al., 2025 (T12) | Feghhi et al., 2025 (T12) | Levin et al., 2026 (T12/T15/T16) | Y. Zhang et al., 2026 (T12/T15) | Nason-Tomaszewski et al., 2026 (T16) | Boccato et al., 2026 (T12/T15/T16/T17) | Jude et al., 2026a (T17/T18) | Jude et al., 2026b (T17) | Yoon et al., 2026 (T12) | Card et al., 2026 (T15) | Fogg et al., 2026 (T12/T15/T16/T17/T21/T22) | Z. Zhang et al., 2026 (T12/T15) | Pouratian et al., 2026 (V1/V2/V3) | Fan et al., 2023 (T5) | Karpowicz et al., 2024 (T5) | Willett et al., 2021 (T5) | Wairagkar et al., 2025 (T15) | Levin et al., 2026 (T5/T12/T18) | Wairagkar et al., 2026 (T15)

Filtered to 8 open-data studies exceeding native-speaker vocabulary (>71,400 word forms)

(b) Open-sourced dataset size

released sentences

10 d/sess, 95 blks, span 253 d: 1,000
24 d/sess, 275 blks, span 120 d: 9,680
21 d/sess, 105 blks, span 287 d: 1,315
45 d/sess, 265 blks, span 612 d: 9,498
37 d/sess, blks n/a, span 171 d: 8,489
1 d/sess, 12 blks, span 1 d: 500
3 d/sess, 18 blks, span 309 d: 720
2 d/sess, 14 blks, span 234 d: 450
8 d/sess, 46 blks, span 89 d: 668
9 d/sess, 98 blks, span 119 d: 1,033
13 d/sess, 96 blks, span 141 d: 3,674

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | Jude et al., 2026a (T17) | Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(c) Word error rate (online)
no series printed: Wairagkar et al., 2025 (listener median)

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

word error rate (%)

12.8% 14-dataset average (Radford et al., 2022)
5.8% Switchboard (Xiong et al., 2017)
2.5% LibriSpeech (Radford et al., 2022)

vocab open mean±SD† 5 of 10 d: 25.1±8.6
vocab 125k mean±SD† 5 of 24 d: 23.9±1.8
vocab 130k mean±SD† 10 of 21 d: 4.0±1.2
vocab 125k mean±SD 14 of 45 d: 4.9±3.3
vocab open listener median 8 of 37 d: 43.8
vocab 50-w mean±SD† 1 of 1 d: 23.7±0.0
vocab 125k mean±SD† 2 of 3 d: 51.3±33.1
vocab 125k mean±SD† 1 of 2 d: 46.6±0.0
vocab 125k mean±SD† 6 of 8 d: 12.5±2.5
vocab 125k mean±SD† 5 of 6 d: 3.4±0.8
vocab 125k mean±SD† 4 of 13 d: 66.5±18.1

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | Jude et al., 2026a (T17) | Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(d) Communication speed (online)
n/r: speed not reported

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

words per minute

160 wpm natural conversation (Card et al., 2024)
51.6 wpm keyboard typing (Dhakal et al., 2018)
36.2 wpm smartphone typing (Palin et al., 2019)
8 wpm point-and-click BCI (Willett et al., 2021)
5–10 wpm eye typing (Hamid & Kristensson, 2024)

vocab open mean±SD† 5 of 10 d: 15.4±2.2
vocab 125k mean±SD‡ 5 of 24 d: 65.3±0.9
vocab 130k mean±SD† 10 of 21 d (10 released): 14.6±2.2
vocab 125k mean±SD 14 of 45 d: 32.5±2.7
vocab 125k mean±SD† 6 of 8 d: 6.8±1.8
vocab 125k mean±SD† 7 of 9 d: 14.0±4.5
vocab 125k mean±SD† 8 of 13 d: 34.1±7.7

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | Jude et al., 2026a (T17) | Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

Our reproduction on a 64 GB RAM / RTX 5060 Ti (16 GB) home PC (8 open-data studies)

(e) Reproduction: acoustic / synthesis stage (no LM), one pair per panel-(c) bar

our reproduction (16 GB home PC) | paper / benchmark anchor

Greedy PER / CER / 1−r (%)

4.7 | 2.8
23.4 | 19.7
seed: 2.65
6.2 | 6.2
10.2 | 9.5
r: 0.69 vs 0.83
31.4 | 17.0
50-w sess.
61.4 | 24.0
3-sess. subset
59.7 | 54.0
2-sess. subset
61.5 | 52.0
best; SrcData
16.5 | 18.6
paper <12%
5.3 | 12.0
40.0 | 34.0

Willett et al., 2021 (T5, raw CER) | Willett et al., 2023 (T12, greedy PER) | Fan et al., 2023 (T5, CORP vo CER) | Card et al., 2024 (T15, greedy PER) | Wairagkar et al., 2025 (T15, 1−mel r ²%) | Kunz et al., 2025 (T12, greedy PER) | Kunz et al., 2025 (T15, greedy PER) | Kunz et al., 2025 (T16, greedy PER) | Jude et al., 2026a (T17, raw CER) | Jude et al., 2026a (T18, raw CER) | Jude et al., 2026b (T17, greedy PER)

(f) Reproduction: full-pipeline WER, one pair per panel-(c) bar

our reproduction (16 GB home PC) | paper / benchmark anchor

Word error rate (%)

bigram 63-tr. pool
10.0 | 3.6
day med. 5-gram
23.6 | 23.8
recal. arm 10 sess.
4.1 | 6.2
41-s pool biGRU-48k
6.2 | 6.7
WER n/r (voice; see e)
50-w sess. gap attr.
94.3 | 24.0
125k gap attr.
78.1 | 54.0
125k gap attr.
82.4 | 52.0
best sess.
11.5 | 15.0
best; paper <7%
5.7 | 7.0
conv. day-68
44.4 | 50.0

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | Jude et al., 2026a (T17) | Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(g) Host RAM walls: official stacks vs. our 64 GB PC

official / naive stack | our consumer-PC stack | 64 GB our host RAM

Peak host RAM (GB)

300 → 19.6 GB (mmap + .idx; 15.3×)
40–60 → 1.46 GB (KenLM trie; 41×)
64+ → 22.5 GB (2.8×) (drop unpruned FST)
56 → 26.9 GB (2.1×) (mmap + lazy FST)

125k 5-gram WFST LM (host RAM; 4 studies) | 125k 3-gram WFST LM (host RAM; 3 studies) | Typing 5-gram WFST (host RAM; Jude 2026a) | Handwriting fold + FST (host RAM; Willett 2021)

(h) GPU VRAM: official rigs vs. our single 16 GB GPU

official provisioned GPU VRAM | our peak VRAM (1× RTX 5060 Ti) | 16 GB our GPU VRAM

Peak GPU VRAM (GB)

80 GB A100 → 13.0 GB (OPT fp16)
24–48 GB 4090 → 13.0 GB (accum + OPT)
3×24 GB 3090 + A6000 → 0–1.5 GB (CPU / .h5)
40–44 GB rig → 3.5–4.0 GB (GRU + GPT-2)
24 GB 4090 → 11.5 GB (bf16 port)

Speech RNN + OPT-6.7B (A100 80G; Willett 2023) | Speech RNN + OPT-6.7B (1–2×4090, 3 studies) | Voice synthesis (3×-3090; Wairagkar 2025) | Handwriting GRU + GPT-2 (1080 Ti / A100; 2 studies) | Typing GRU + 5-gram (RTX 4090; Jude 2026a)

(i) Session-level PER vs. post-LM WER

word error rate after LM stage (%)

No-LM baseline: 1 − (1 − PER)^3.4
WER = PER
PER = 25%
(≤25%: 45/57 = 79% WER < PER;
>25%: 4/20 = 20% WER < PER)
This means that we must get PER < 25%.
per-session (open) pooled (star)
Willett et al., 2023 (T12)
Card et al., 2024 (T15)
Jude et al., 2026b (T17)

greedy phoneme error rate per session (%)